

Separable Hilbert spaces need not have the nice-geodesics structure

Full counterexample for arXiv:2204.01461

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Status: full negative answer. Every infinite-dimensional separable Hilbert space fails the source's nice-geodesics condition, for every choice of countable dense test set.

Source question

Let H be a separable Hadamard space and let $(y_n)_{n \in \mathbb{N}}$ be dense in H . The paper calls the geodesic structure *nice* if, for every $x \in H$ and every sequence (x_k) ,

$$P_{[x, y_n]} x_k \longrightarrow x \quad \text{for every } n \implies P_\gamma x_k \longrightarrow x \quad \text{for every geodesic segment } \gamma \text{ starting at } x.$$

After using this as a hypothesis in its compactness theorem, the paper asks: “Does every separable Hadamard space enjoy the nice geodesics structure?”

Counterexample theorem

Theorem 1. *Let H be an infinite-dimensional separable real Hilbert space. For every countable dense sequence (y_n) in H , there exist a unit vector u and an unbounded sequence (z_k) such that*

$$P_{[0, y_n]} z_k \longrightarrow 0 \quad (n \in \mathbb{N}), \quad P_{[0, u]} z_k = u \quad (k \in \mathbb{N}).$$

Consequently H , and in particular ℓ^2 , does not have the nice-geodesics structure.

Proof. Set $E_k = \text{span}\{y_1, \dots, y_k\}$. Each E_k is a finite-dimensional closed nowhere-dense subset of H . By the Baire category theorem, choose a unit vector

$$u \notin \bigcup_{k=1}^{\infty} E_k.$$

Let Q_k be the orthogonal projection onto E_k , put $v_k = u - Q_k u \neq 0$, and define

$$z_k = \frac{v_k}{\|v_k\|^2}.$$

For $n \leq k$, we have $z_k \perp E_k$ and hence $\langle z_k, y_n \rangle = 0$. The nearest point to z_k on the segment $[0, y_n] = \{ty_n : 0 \leq t \leq 1\}$ is therefore exactly 0. Thus, for each fixed n , $P_{[0, y_n]} z_k = 0$ for every $k \geq n$.

On the other hand, $v_k \perp Q_k u$, so

$$\langle z_k, u \rangle = \frac{\langle v_k, Q_k u + v_k \rangle}{\|v_k\|^2} = 1.$$

For $0 \leq t \leq 1$,

$$\|z_k - tu\|^2 = \|z_k\|^2 - 2t + t^2,$$

whose unique minimum occurs at $t = 1$. Hence $P_{[0, u]} z_k = u$ for every k , contradicting the defining implication.

Finally, density of (y_n) makes $\bigcup_k E_k$ dense in H , whence $Q_k u \rightarrow u$. Thus $\|v_k\| \rightarrow 0$ and $\|z_k\| = \|v_k\|^{-1} \rightarrow \infty$. $\square$

Sharp Hilbert-space classification

Theorem 2. *Every finite-dimensional Hilbert space has the nice-geodesics structure, for every countable dense test set.*

Proof. Translate the base point to 0. Suppose $P_{[0,y_n]}z_k \rightarrow 0$ for every n . First, (z_k) is bounded. Otherwise, after passing to a subsequence, compactness of the unit sphere gives $z_k/\|z_k\| \rightarrow w$ for a unit vector w . Choose a test point y_n close enough to w that $\langle w, y_n \rangle > 0$. Then $\langle z_k, y_n \rangle \rightarrow +\infty$, so eventually the projection of z_k onto $[0, y_n]$ is the nonzero endpoint y_n , a contradiction.

Let $M = \sup_k \|z_k\|$. For a nonzero y , the projection onto $[0, y]$ is

$$P_{[0,y]}z = \min \left\{ 1, \max \left\{ 0, \frac{\langle z, y \rangle}{\|y\|^2} \right\} \right\} y.$$

On the compact set $\{z : \|z\| \leq M\}$ this expression depends uniformly continuously on y near any fixed nonzero endpoint. Approximate an arbitrary y by one fixed test point y_n , then let $k \rightarrow \infty$. It follows that $P_{[0,y]}z_k \rightarrow 0$. The zero endpoint is trivial, so the defining implication holds for every segment. $\square$

Corollary 1. *Among separable Hilbert spaces, the nice-geodesics property is equivalent to finite dimensionality.*

Scope and priority check

Theorem 1 completely answers the source's universal question in the negative; no additional curvature or nonlinear construction is needed. It does not decide the paper's separate question about weak compactness versus weak sequential compactness for arbitrary unbounded sets.

The run indexes and bounded primary-source searches through 13 August 2026 found no later answer under the paper's terminology. The closely related same-day paper arXiv:2204.01291 analyzes weak topologies for unbounded nets and the nonsequential weak topology of infinite-dimensional Hilbert space, but it does not state this countable-segment counterexample. Priority should still be checked by a specialist before external dissemination.